

SRG run: example_water_nacl_davies

SRG Chemical Computing Platform v6.1

May 3, 2026

1 Experimental settings

Run identifier (escaped): example_water_nacl_davies.

=== electrolyte_params_report.json ===

```
{
  "settings": {
    "model": "davies",
    "use_onsager": true,
    "box_volume_l": 1e-22,
    "temperature_K": 298.15,
    "dielectric": 78.54,
    "ion_size_nm": 0.35,
    "pitzer_beta0": 0.0765,
    "pitzer_beta1": 0.2664,
    "pitzer_cphi": 0.00127,
    "pitzer_alpha": 2.0,
    "pitzer_b": 1.2,
    "lambda0_s_cm2_mol": 126.45,
    "onsager_K": 60.0
  },
  "warning": "Debye-Huckel/Davies are dilute-solution approximations; Pitzer here is a",
  "validity_hint": "Use dh/extended_dh qualitatively at low ionic strength; Davies oft",
}
```

Table 1: Summary metrics from the SRG simulation run.

Quantity	Value
Initial energy	-157.4
Final energy	-346.2
Minimum energy	-346.7
Final absorbance proxy	0.04709
Final ionic strength (mol per liter)	0.09963
Final Debye length (nm)	0.964
Final gamma mean	0.7817
Final conductivity proxy (S per cm)	0.01071
Final Dirac gap	1.145
Final braid reduced length	215

2 Summary metrics

3 Results narrative

Results summary. Across 220 simulation steps (example_water_nacl_davies), the coarse-grained free energy evolved from -157.4 to -346.2 (minimum -346.7). The electrolyte layer reports ionic strength $I = 0.09963$ (mol per liter) and Debye screening length $\lambda_D = 0.964$ nm. Spectral indicators yield a Dirac gap of 1.145 (simulator units). Braid complexity (reduced word length) reaches 215. These metrics are qualitative graph-level proxies; quantitative agreement with experiment is not implied.

4 Figures

5 Interpretation

The traces summarize coarse-grained graph energetics, optional electrolyte corrections, spectral indicators from the discrete Dirac operator, and braid statistics on Monte Carlo moves.

6 Limitations

This MVP uses qualitative energy mappings and simplified DH/Davies/Pitzer proxies; it is not a calibrated molecular simulation or experimental predictor.

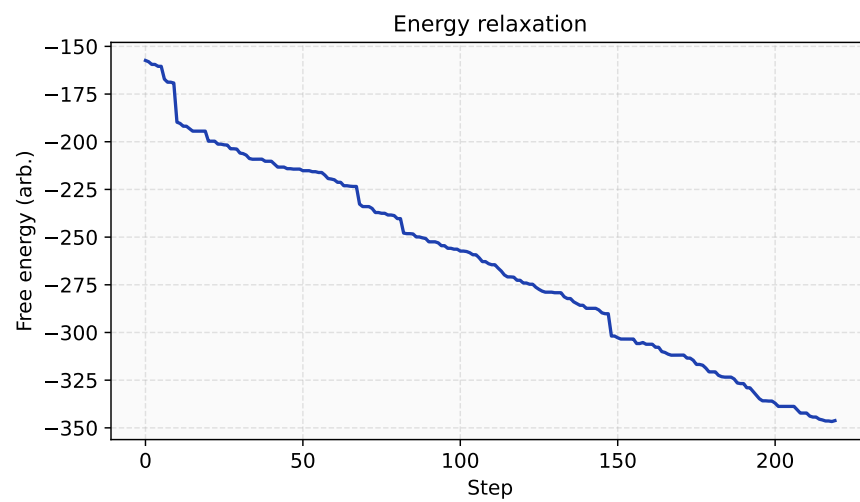


Figure 1: Figure: fig_energy

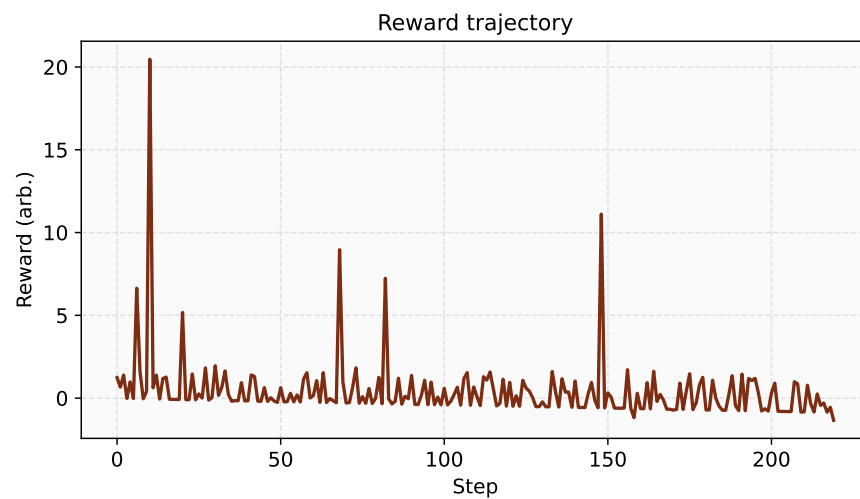


Figure 2: Figure: fig_reward

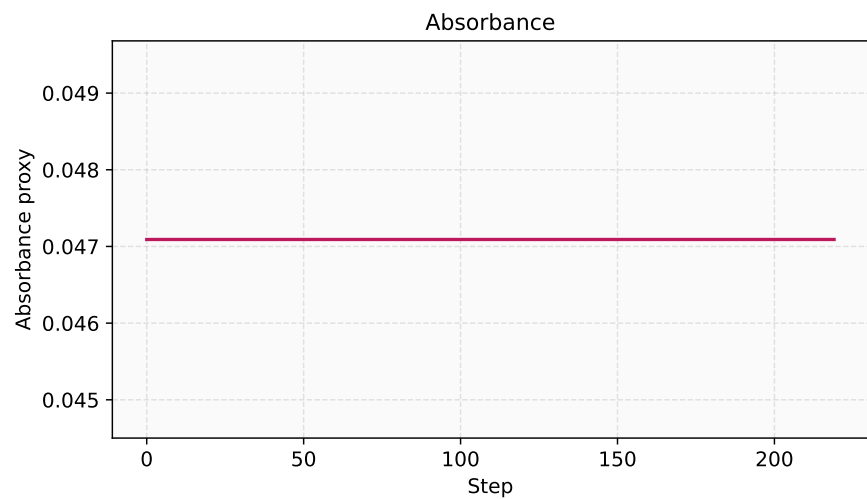


Figure 3: Figure: fig_absorbance

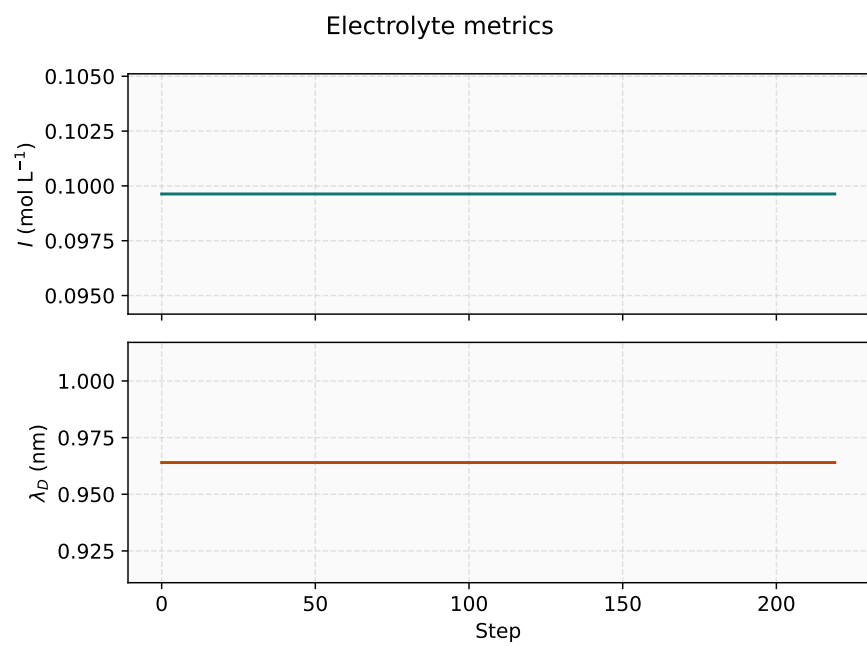


Figure 4: Figure: fig_electrolyte

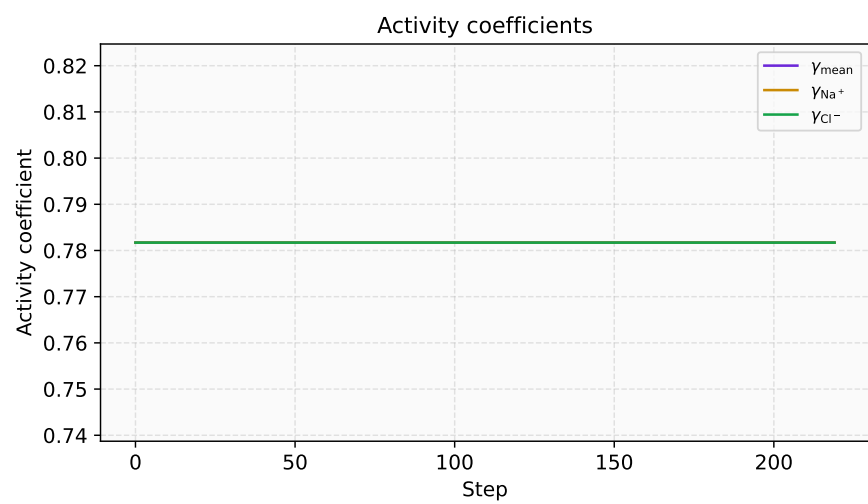


Figure 5: Figure: fig_activity

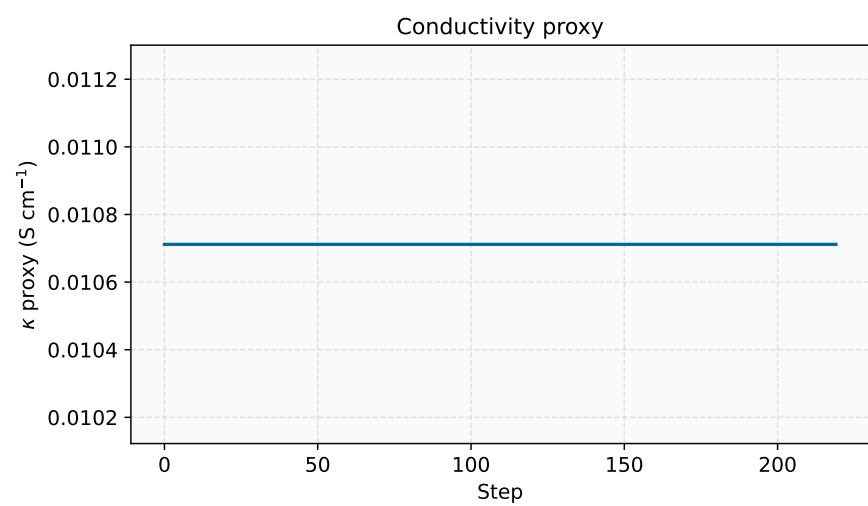


Figure 6: Figure: fig_conductivity

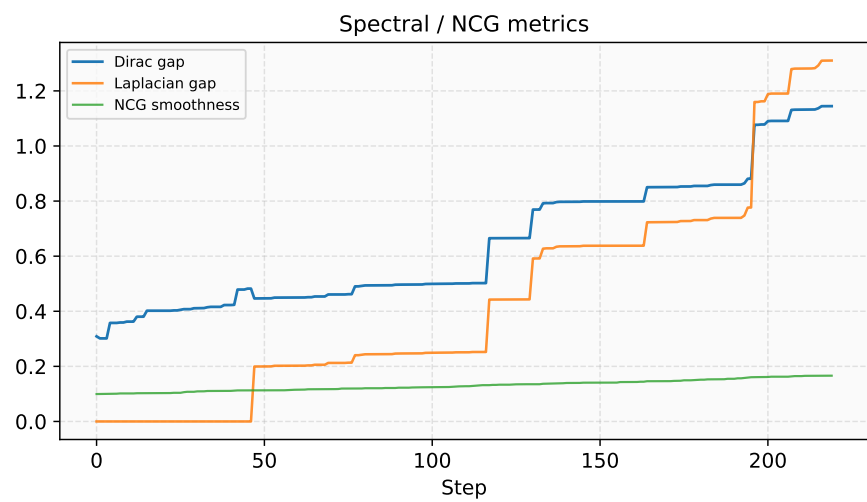


Figure 7: Figure: fig_spectral

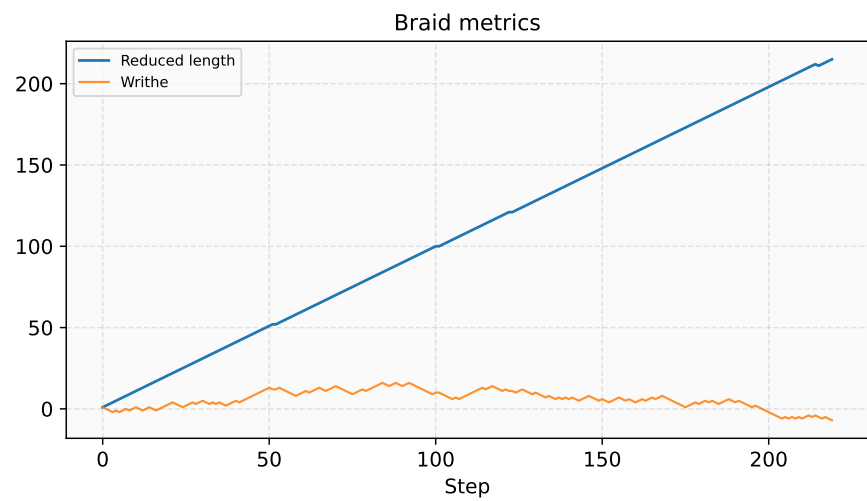


Figure 8: Figure: fig_braid

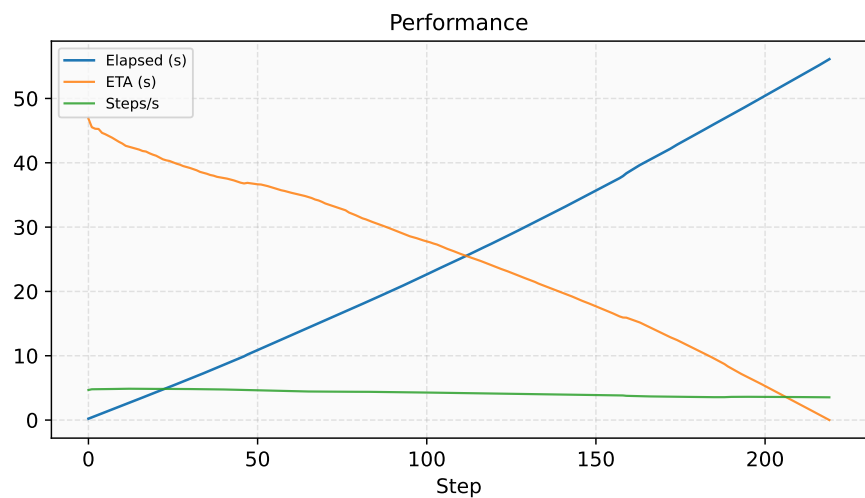


Figure 9: Figure: fig_performance

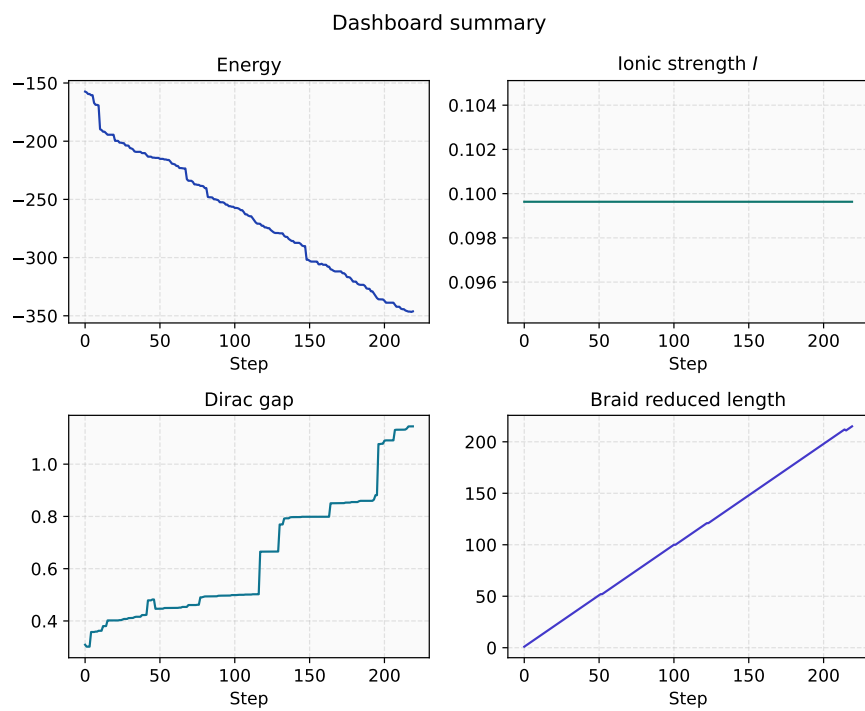


Figure 10: Figure: fig_dashboard_summary